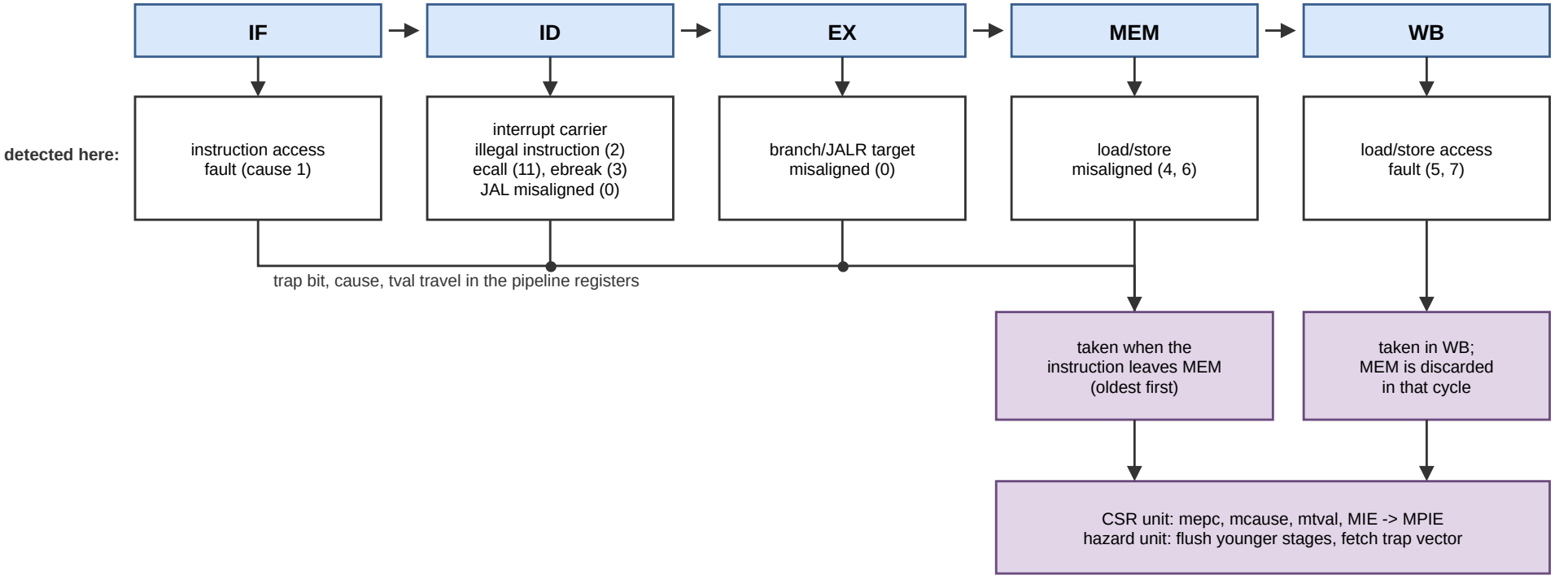
IF
ID
EX
MEM
WB
detected here:
instruction access fault (cause 1)
interrupt carrier
illegal instruction (2)
ecall (11), ebreak (3)
JAL misaligned (0)
branch/JALR target misaligned (0)
load/store misaligned (4, 6)
load/store access fault (5, 7)
trap bit, cause, tval travel in the pipeline registers
taken when the instruction leaves MEM (oldest first)
taken in WB; MEM is discarded in that cycle
CSR unit: mepc, mcause, mtval, MIE -> MPIE
hazard unit: flush younger stages, fetch trap vector